

Cross-fitting-free DML with multiway dependence

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(Preliminary—comments welcome)

Abstract

- ▶ Under **multiway dependence**, we show that **double/debiased machine learning (DML)** estimator is asymptotically normal **without cross-fitting**.
- ▶ We extend arguments for i.i.d. sampling to the multiway dependence
 - ⇒ requires controlling the empirical processes arising from learning nuisance parameter
 - ⇒ we develop **new maximal inequalities** for empirical processes with multiway dependence that may be of independent interest.

Background and Motivation

Simulation Evidence

Asymptotics of DML without Cross-Fitting

Maximal Inequalities under Multiway Cluster Dependence

Cluster dependence in real data environments

- ▶ Real economic data often exhibit **complex correlation structures**, making the i.i.d. assumption invalid.
- ▶ Common multi-way clustering dependencies include
 - ▶ **Regional economic** data: region–industry–time
 - ▶ **Firm-level panel** data: firm–market–year
 - ▶ **International trade** data: exporter–importer–time

Challenges for DML in real data environments

- ▶ Consequences
 - ▶ ML-based inference methods built on independence **lack rigorous theoretical guarantees**.
 - ▶ **overfitting bias** and **over/under rejections**
 - ▶ DML based on special **cross-fitting** algorithms are computationally costly, less accurate, too conservative, very specific about sampling assumptions.

Double Machine Learning

- ▶ CCDDHNR (2018) propose a general methodology for estimation & inference for two-step estimation problems

DML $\approx$ Neyman Orthogonal Score + Cross-Fitting.

- ▶ The former mitigates the bias coming from slow convergence rates of ML-based learning of nuisance parameters.
- ▶ The latter simplifies the derivations and, according to conventional wisdom, may help reduce the bias induced by overfitting (lacks justification outside certain special cases)

A Generic DML Problem

- ▶ A DML problem defined by

$$\mathbb{E}[\psi(W; \theta_0, \eta_0)] = 0$$

where

- ▶ W : data r.v.
- ▶ θ_0 : (scalar) parameter of interest, exactly identified¹
- ▶ η_0 : unknown nuisance parameter (potentially nonparametric/high-dimensional)
- ▶ ψ : score, known up to the parameters, satisfies the *Neyman Orthogonality*

$$\partial_{\eta} \mathbb{E}[\psi(W, \theta_0, \eta_0)](\tilde{\eta}) := \partial_t \mathbb{E}[\psi(W, \theta_0, \eta_0 + t\tilde{\eta})]|_{t=0} = 0, \quad \text{for all } \tilde{\eta} \in \Gamma$$

- ▶ An estimator $\hat{\eta} = \hat{\eta}(\{W_i\}_{i \in I})$ for η_0 is available

¹Over-identification case is also studied in the paper.

Example 1: Partially Linear Model

- ▶ Consider the partially linear model (Robinson, 1988)

$$Y = \theta D + m_0(X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D, X] = 0$$

and the variable of interest D admits a first stage projection

$$D = g_0(X) + \nu, \quad \mathbb{E}[\nu \mid X] = 0$$

- ▶ Write $\eta_0 = (m_0, g_0)$ and $W = (Y, D, X')'$

$$\psi(W; \theta, \eta) = (Y - \theta D - m(X)) \cdot (D - g(X))$$

- ▶ Can verify that

$$\mathbb{E}[\psi(W; \theta_0, \eta_0)] = 0$$

and the Neyman orthogonality condition

Example 2: AIPW Estimator

- ▶ Assume unconfoundedness and overlap:

$$(Y(1), Y(0)) \perp D \mid X, \quad 0 < \Pr(D = 1 \mid X) < 1,$$

- ▶ The parameter of interest is the average treatment effect

$$\theta_0 = \mathbb{E}[Y(1) - Y(0)].$$

- ▶ Define the nuisance functions

$$\mu_{10}(X) = \mathbb{E}[Y \mid D = 1, X], \quad \mu_{00}(X) = \mathbb{E}[Y \mid D = 0, X],$$

- ▶ Can verify the AIPW score is Neyman orthogonal

$$\psi(W; \theta, \eta) = \mu_1(X) - \mu_0(X) + \frac{D}{e(X)} \{Y - \mu_1(X)\} - \frac{1 - D}{1 - e(X)} \{Y - \mu_0(X)\} - \theta.$$

Example 3: DiD Analysis through DML

- ▶ Callaway & Sant'Anna's (2021) doubly-robust group-time (g and t) ATT estimand: for $t > g - \delta$, $\delta > 0$ is a known no-anticipation buffer window:

$$\text{ATT}(g, t; \delta) = \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{\mathbb{E} \left[\frac{p_g(X)C}{1-p_g(X)} \right]} \right) \times (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}(X)) \right]$$

- ▶ G_g is the indicator for the group membership and C is the indicator for the never-treated group.
- ▶ Nuisance functions:
 - ▶ $p_g(X) = P(G_g = 1 \mid X, G_g + C = 1)$: generalized propensity score
 - ▶ $m_{g,t,\delta}(X) = \mathbb{E}[Y_t - Y_{g-\delta-1} \mid X, C = 1]$: outcome regression
- ▶ The corresponding score is Neyman orthogonal w.r.t. p_g and $m_{g,t,\delta}$.

DML with Cross-Fitting

Score Nuisance



$$\mathbb{E}_{n,1}[\psi(W; \theta, \hat{\eta}_1)]$$

Nuisance Score



$$\mathbb{E}_{n,2}[\psi(W; \theta, \hat{\eta}_2)]$$

DML with Cross-Fitting

- ▶ Randomly partition $\{1, \dots, n\}$ into K splits $\{I_1, \dots, I_K\}$
- ▶ For each $s \in \{1, \dots, K\}$, obtain an estimate

$$\hat{\eta}_s = \hat{\eta}((W_i)_{i \in \{1, \dots, N\} \setminus I_s})$$

using only the subsample with $i \in \{1, \dots, N\} \setminus I_s$

- ▶ Define $\tilde{\theta}$, a DML estimator for θ_0 , as the solution to

$$\sum_{s=1}^S \mathbb{E}_{n,s}[\psi(W; \theta, \hat{\eta}_s)] = 0,$$

where $\mathbb{E}_{n,s}[f(W)] = |I_s|^{-1} \sum_{i \in I_s} f(W_i)$ denotes the subsample empirical mean using only data with $i \in I_s$

Multiway Dependence

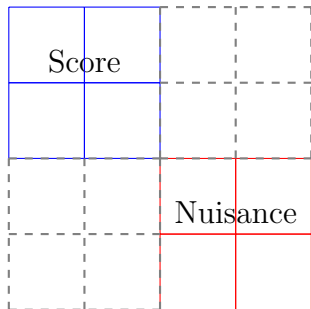
- ▶ Suppose the data follow two-way dependence:

$$W_{i_1 i_2} \perp W_{j_1 j_2}$$

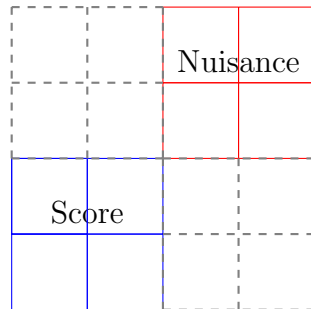
unless $i_1 = j_1$ or $i_2 = j_2$

- ▶ Examples include
 - ▶ market share data ($i_1 = \text{market}, i_2 = \text{product}$)
 - ▶ scanner data ($i_1 = \text{store}, i_2 = \text{product}$)
 - ▶ bipartite network data ($i_1 = \text{book}, i_2 = \text{reader}$)
 - ▶ matched employer-employee data ($i_1 = \text{employer}, i_2 = \text{employee}$)
 - ▶ matched student-teacher data ($i_1 = \text{s}, i_2 = \text{t}$)
- ...
- ▶ Chiang, Kato, Ma, Sasaki (2023): multiway cross-fitting

DML with Multiway Cross-Fitting

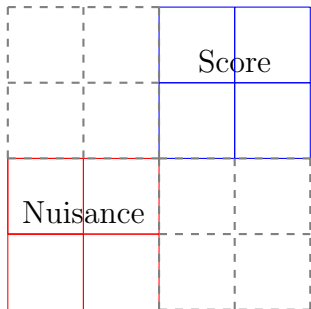


$$\mathbb{E}_{n,11}[\psi(\textcolor{blue}{W}; \theta, \hat{\eta}_{11})]$$

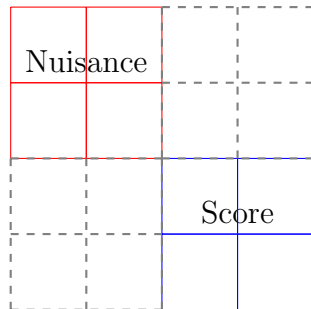


$$\mathbb{E}_{n,21}[\psi(\textcolor{blue}{W}; \theta, \hat{\eta}_{21})]$$

DML with Multiway Cross-Fitting



$$\mathbb{E}_{n,12}[\psi(\textcolor{blue}{W}; \theta, \hat{\eta}_{\textcolor{red}{12}})]$$



$$\mathbb{E}_{n,22}[\psi(\textcolor{blue}{W}; \theta, \hat{\eta}_{\textcolor{red}{22}})]$$

DML with Multiway Cross-Fitting

- ▶ Dataset $\{W_{i_1 i_2} : 1 \leq i_1 \leq N_1, 1 \leq i_2 \leq N_2\}$
- ▶ Randomly partition $\{1, \dots, N_1\}$ into $\{I_1, \dots, I_K\}$ and $\{1, \dots, N_2\}$ into $\{J_1, \dots, J_K\}$
- ▶ For each $(s, t) \in \{1, \dots, K\}^2$, obtain a nuisance parameter estimate

$$\hat{\eta}_{st} = \hat{\eta} \left((W_{i_1 i_2})_{(i_1, i_2) \in (\{1, \dots, N_1\} \setminus I_s) \times (\{1, \dots, N_2\} \setminus J_t)} \right)$$

- ▶ Multiway DML estimator $\tilde{\theta}$ is obtained by solving

$$\sum_{s=1}^S \sum_{t=1}^S \mathbb{E}_{n, st} [\psi(W; \theta, \hat{\eta}_{st})] = 0,$$

where $\mathbb{E}_{n, st}[f(W)] = (|I_s| |J_t|)^{-1} \sum_{(i_1, i_2) \in I_s \times J_t} f(W_{i_1 i_2})$

Issues with Cross-Fitting

- ▶ Estimator is random given the data, especially for small K
- ▶ Nuisance parameter is poorly estimated, especially when K is small
- ▶ Computationally demanding, especially for large K or complicated first stage
 - ⇒ these difficulties are exacerbated under multiway dependence
- ▶ Wider confidence intervals

Illustrating Example of Nuisance Estimation

- ▶ Consider $\eta_0 = \mathbb{E}[W_{i_1 i_2}]$ and

$$W_{i_1 i_2} = a_{i_1} + \varepsilon_{i_1 i_2}, \quad a_{i_1}, \varepsilon_{i_1 i_2} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$$

- ▶ Suppose $n = N_1 = N_2$, for sample average estimator

- ▶ **Full-sample estimator :**

$$\text{Var}(\hat{\eta}_{\text{full}}) = \frac{1}{n} + \frac{1}{n^2} \sim \frac{1}{n}$$

- ▶ **2²-fold cross-fitted estimator:**

$$\text{Var}(\hat{\eta}_{st}) = \frac{2}{n} + \frac{4}{n^2} \sim \frac{2}{n}$$

Cross-Fitting-Free DML

- ▶ Sometimes (doubly) cross-fitting is beneficial: Newey and Robins (2018) – “complex” first stage, large sample
- ▶ Classical semiparametrics avoid cross-fitting using stochastic equicontinuity:
 - ▶ Robinson (1988), Ichimura (1993), Andrews (1994), ...etc
- ▶ DML without cross-fitting in special settings: Belloni, Chernozhukov, Hansen (2012), Farrell (2015), Belloni et al (2016), Chen (2025)
- ▶ Recent DML literature also avoids cross-fitting with new techniques:
 - ▶ i.i.d. sampling: Belloni, Chernozhukov, Kato (2015), Chen, Syrgkanis, Austern (2022)
 - ▶ Weak dependence (β -mixing): Cao and Leung (2025)
 - ▶ Multiway dependence: **THIS PAPER**

Background and Motivation

Simulation Evidence

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- ▶ A partial linear IV model:

$$\begin{aligned}Y_{ij} &= \theta D_{ij} + g(X_{ij}) + U_{ij}, \quad \theta = 1 \\g(X_{ij}) &= \sin(X_{ij,1} + X_{ij,2}) + 0.5X_{ij,3} + \cdots + 0.5X_{ij,d} \\D_{ij} &= 0.5Z_{ij} + 0.2(X_{ij,1} + X_{ij,2})^2 + V_{ij} \\Z_{ij} &= X_{ij,2}X_{ij,3} + \varepsilon_{ij} \\X_{ij,k} &= \alpha_{i,k} + \gamma_{j,k} + \epsilon_{ij,k}, \quad k = 1, \dots, d \\U_{ij} &= \alpha_{i,u} + \gamma_{j,u} + \epsilon_{ij,u} \\V_{ij} &= \alpha_{i,v} + \gamma_{j,v} + \epsilon_{ij,v} \\\varepsilon_{ij} &= \alpha_{i,\varepsilon} + \gamma_{j,\varepsilon} + \epsilon_{ij,\varepsilon}\end{aligned}$$

Machine Learners Tuning

Table: Tuning choices for nuisance-function learners

Learner	Features	Main tuning choices	Tuning rule
Ridge spline	cubic natural spline, 3 knots, raw dimension 5^d	5-fold CV	<code>lambda.min</code>
LASSO spline, MPB	Same spline basis	Initial $\tilde{\lambda} = \log(n)\sqrt{\log(p)/n}$; MPB with $B = 200$, $\alpha = 0.10$, $c = 1.10$; final $\lambda = 2cq_{1-\alpha}$	fixed
Elastic net spline	Same spline basis	$\alpha = 0.50$; 5-fold CV	<code>lambda.min</code>
Random forest	Raw X	500 trees; $mtry = \max\{1, \lfloor \sqrt{d_X} \rfloor\}$; min node size 5	fixed
Neural net	Raw X	Dense network (64, 32) with ReLU; adaptive SGD (Adam) with learning rate 10^{-3} ; 30 epochs; batch size 32; validation split 0.2 with MSE loss and patience 5 for early stop.	fixed

Notes: Each learner is used separately for the nuisance functions of Y , D , and Z . For spline learners, knots, centering, and scaling are computed using the training sample.

$N = M$	d_X	K	Learner	Bias	SD	RMSE	Coverage	Runtime/MPB	CF/No-CF
50	4	0	MPB LASSO	0.014	0.149	0.150	0.933	$1.00\times$	Ref.
			Ridge	0.000	0.144	0.143	0.930	$10.41\times$	Ref.
			Elastic net	-0.005	0.144	0.144	0.932	$29.25\times$	Ref.
			Random forest	-0.023	0.125	0.127	0.933	$5.20\times$	Ref.
			Neural network	0.016	0.146	0.147	0.926	$26.63\times$	Ref.
50	4	2	MPB LASSO	0.007	1.585	1.584	0.924	$1.00\times$	$1.12\times$
			Ridge	-0.033	0.118	0.122	0.966	$4.62\times$	$0.50\times$
			Elastic net	-0.059	0.421	0.425	0.958	$8.78\times$	$0.34\times$
			Random forest	-0.027	0.105	0.109	0.973	$3.77\times$	$0.81\times$
			Neural network	0.023	0.126	0.128	0.937	$66.83\times$	$2.82\times$
50	4	4	MPB LASSO	-0.003	0.167	0.167	0.990	$1.00\times$	$7.17\times$
			Ridge	-0.023	0.150	0.152	0.974	$11.36\times$	$7.82\times$
			Elastic net	-0.038	0.293	0.296	0.986	$40.25\times$	$9.87\times$
			Random forest	-0.025	0.116	0.119	0.988	$5.52\times$	$7.61\times$
			Neural network	0.026	0.134	0.136	0.977	$95.23\times$	$25.65\times$

$N = M$	d_X	K	Learner	Bias	SD	RMSE	Coverage	Runtime/MPB	CF/No-CF
25	4	0	MPB LASSO	0.086	0.265	0.278	0.899	1.00×	Ref.
			Ridge	-0.019	0.165	0.166	0.949	3.82×	Ref.
			Elastic net	0.001	0.223	0.223	0.920	6.85×	Ref.
			Random forest	-0.035	0.151	0.155	0.952	3.22×	Ref.
			Neural network	0.033	0.196	0.198	0.897	42.23×	Ref.
25	4	2	MPB LASSO	0.075	1.047	1.049	0.954	1.00×	1.67×
			Ridge	-0.002	0.569	0.569	0.921	3.53×	1.54×
			Elastic net	0.119	13.084	13.078	0.899	2.34×	0.57×
			Random forest	-0.027	0.122	0.125	0.974	1.91×	0.99×
			Neural network	0.041	0.211	0.215	0.941	108.49×	4.28×
25	4	4	MPB LASSO	-0.334	8.643	8.645	0.984	1.00×	8.87×
			Ridge	-0.031	0.177	0.179	0.977	4.21×	9.78×
			Elastic net	-0.071	1.680	1.680	0.975	3.03×	3.93×
			Random forest	-0.031	0.137	0.140	0.996	2.63×	7.25×
			Neural network	0.030	0.176	0.178	0.990	182.14×	38.27×

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Asymptotics of DML

- ▶ Let's derive asymptotics of $\hat{\theta}$ together!
- ▶ Suppose the data $\{W_i\}_{i=1}^n$ are i.i.d. copies of W
- ▶ Given a nuisance parameter estimator $\hat{\eta}$, an estimator $\hat{\theta}$ for θ_0 can be defined as the solution of

$$\mathbb{E}_n[\psi(W; \theta, \hat{\eta})] := \frac{1}{n} \sum_{i=1}^n \psi(W_i; \theta, \hat{\eta}) = 0$$

- By definition and linearisation

$$\begin{aligned}
 0 &= \mathbb{E}_n[\psi(W; \hat{\theta}, \hat{\eta})] \\
 &= \mathbb{E}_n[\psi(W; \theta_0, \hat{\eta})] + \underbrace{\mathbb{E}_n[\partial_{\theta}\psi(W; \theta_0, \hat{\eta})]}_{(\text{assume } \neq 0)}(\hat{\theta} - \theta_0) \\
 &\quad + O_p(\|\hat{\theta} - \theta_0\|^2)
 \end{aligned}$$

- Thus, with consistency of $\hat{\theta}$ and $\|\hat{\eta} - \eta_0\|_{P,2} = o_p(n^{-1/4})^2$, then

$$\begin{aligned}
 \hat{\theta} - \theta_0 &= - \underbrace{(\mathbb{E}_n[\partial_{\theta}\psi(W; \theta_0, \hat{\eta})])^{-1}}_{\xrightarrow{P} \mathbb{E}[\partial_{\theta}\psi(W; \theta_0, \eta_0)]^{-1}} \mathbb{E}_n[\psi(W; \theta_0, \hat{\eta})] \\
 &\quad + o_p(n^{-1/2})
 \end{aligned}$$

- It suffices to establish asymptotics for

$$\mathbb{E}_n[\psi(W; \theta_0, \hat{\eta})] = (*)$$

² $\|f\|_{P,2} = \sqrt{\mathbb{E}[f(W)^2]}$ is the $L^2(P)$ -norm.

Bias & Empirical Process Decomposition

- ▶ Recall that $\mathbb{E}[\psi(W; \theta_0, \eta_0)] = 0$
- ▶ By adding and subtracting terms

$$(*) = \underbrace{\mathbb{E}_n[\psi(W; \theta_0, \eta_0)]}_{=I} + \underbrace{\mathbb{E}[\psi(W; \theta_0, \hat{\eta})]}_{=II} + \underbrace{n^{-1/2} \mathbb{G}_n[\psi(W; \theta_0, \hat{\eta}) - \psi(W; \theta_0, \eta_0)]}_{=III}$$

where $\mathbb{G}_n(f) = n^{-1/2} \sum_{i=1}^n (f(W_i) - \mathbb{E}[f(W_i)])$ for a given f

Limiting Distribution (Standard)

- ▶ Term I consists of i.i.d. centred summands, by a standard CLT

$$\sqrt{n}I = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(W_i; \theta_0, \eta_0) \xrightarrow{d} \mathcal{N}(0, \text{Var}(\psi(W; \theta_0, \eta_0)))$$

- ▶ For non-independent cases, so long as a CLT is available, term I remains asymptotically normal
 - $\Rightarrow$ Under multiway dependence, CLTs are available from, e.g. Davezies, D'Haultfoeuille, Guyonvarch (2021), Chiang, Kato, Sasaki (2023), ...

Controlling Bias (Standard)

- ▶ II is a bias term in which the dependence does NOT play a direct role
- ▶ Thanks to orthogonality and regularity conditions,

$$\begin{aligned} & \mathbb{E}[\psi(W; \theta_0, \hat{\eta})] \\ &= \mathbb{E}[\psi(W; \theta_0, \eta_0)] + \partial_{\eta} \mathbb{E}[\psi(W; \theta_0, \eta_0)](\hat{\eta}) + O(\|\hat{\eta} - \eta_0\|_{P,2}^2) \\ &= 0 + 0 + O(\|\hat{\eta} - \eta_0\|_{P,2}^2) \end{aligned}$$

- ▶ If $\|\hat{\eta} - \eta_0\|_{P,2} = o_p(n^{-1/4})$, then $\sqrt{n}|II| = o(1)$

Controlling Empirical Process with Cross-Fitting

- ▶ Suppose $\hat{\eta} \perp \{W_i\}_{i=1}^n$
- ▶ Conditionally on $\hat{\eta}$, its summands are i.i.d. with zero mean.

$$\begin{aligned} n^{-1/2}|III| &= |\mathbb{G}_n[\psi(W; \theta_0, \hat{\eta}) - \psi(W; \theta_0, \eta_0)]| \\ &= \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\{ \psi(W_i; \theta_0, \hat{\eta}) - \psi(W_i; \theta_0, \eta_0) - \mathbb{E}[\psi(W_i; \theta_0, \hat{\eta}) - \psi(W_i; \theta_0, \eta_0)] \right\} \right| \\ &\lesssim_p \sqrt{\mathbb{E} \left[\left(\psi(W; \theta_0, \hat{\eta}) - \psi(W; \theta_0, \eta_0) \right)^2 \right]} \lesssim_p \|\hat{\eta} - \eta_0\|_{P,2}^\nu \end{aligned}$$

for some $\nu > 0$, under regularity conditions

$$\Rightarrow \|\hat{\eta} - \eta_0\|_{P,2} = o_p(1) \text{ suffices for } \sqrt{n}|III| = o_p(1)$$

Controlling Empirical Process without Cross-Fitting

- ▶ Without cross-fitting: $\hat{\eta} \not\perp \{W_i\}_{i=1}^n$
- ▶ Suppose for some $\mathcal{H}_n \ni \eta_0$, we have $\hat{\eta} \in \mathcal{H}_n$ with probability $1 - o(1)$. Define

$$\mathcal{F}_n = \{w \mapsto \psi(w; \theta_0, \eta) - \psi(w; \theta_0, \eta_0) : \eta \in \mathcal{H}_n\}$$

- ▶ Under i.i.d., $\mathbb{G}_n(f)$, $f \in \mathcal{F}_n$ consists of i.i.d. summands
- ▶ Can apply a maximal inequality to control these “localised” empirical processes

$$\sqrt{n}|\mathbb{G}_n(f)| \lesssim_p \sup_{f \in \mathcal{F}_n} |\mathbb{G}_n(f)| \lesssim_p \frac{\{\text{”Size” \& ”Complexity” of } \mathcal{F}_n\}}{\sqrt{n}}$$

- ▶ Similar to the classical semiparametrics (relies on stochastic equicontinuity over a fixed $\mathcal{F}$)

Localisation of Empirical Process

- ▶ Machine learners have growing complexities
 $\Rightarrow$ tackle the growing "Complexity" with shrinking "Size"
- ▶ "Size" of $\mathcal{H}_n$ (hence $\mathcal{F}_n$) can be chosen to contract to η_0 sharply if a good rate of $\|\hat{\eta} - \eta_0\|_{P,2}$ is known
 $\Rightarrow$ typically so since the rate is needed for orthogonality
- ▶ $\sqrt{n}|III| = o_p(1)$ so long as the "Complexity" do not grow too fast with n

$$\{\text{"Size" \& "Complexity" of } \mathcal{F}_n\} = o(\sqrt{n})$$

can be obtained by using a maximal inequality

- ▶ However, such a maximal inequality is **NOT available** under **multiway dependence**
 $\Rightarrow$ we develop our own

Summary & What Else Is in This Paper?

- ▶ Multiway DML without Cross-Fitting:
Localisation → **Hoeffding-Type Decomposition** → **Maximal Inequalities**
- ▶ What else is in the paper?
 - ▶ Generic **debiased GMM** estimators under general multiway dependence
 - ▶ Asymptotic normality holds **without** cross-fitting under general high-level complexity conditions (VC-type) as well as smoothness boundedness of the score function.
 - ▶ High-level conditions are satisfied for some **regularised GLMs** (p and sparsity), **tree-based learners** (p and leaves), and **neural networks** (p , layers, hidden units, total parameters)
 - ▶ **Multiway-cluster robust** consistent variance estimator

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Maximal Inequalities under Multiway Cluster Dependence

Complexity

A class $\mathcal{F}$ is of VC-type if for an envelope F^3 and some positive characteristics (A, v)

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q,2}, \varepsilon \|F\|_{Q,2}) \leq \left(\frac{A}{\varepsilon}\right)^v \quad \text{for all } 0 < \varepsilon \leq 1$$

- ▶ $N(T, \|\cdot\|, \varepsilon)$: covering number for a set T w.r.t. a pseudonorm $\|\cdot\|$, $\varepsilon > 0$
- ▶ Supremum over all finite discrete measures Q
- ▶ Examples: a finite class, a class of indicator functions, a class indexed by a finite-dimensional vector space, a class of smooth functions indexed by finite-dimensional parameters, their Lipschitz and/or bounded-variation transformations, and finite union/intersection/sum/product/division of these classes...

³i.e. $|f(x)| \leq F(x)$ for all x and $f \in \mathcal{F}$.

Maximal Inequality under i.i.d.

- ▶ Suppose $\{W_i\}_{i=1}^n$ are i.i.d. From the literature (Pollard, 1990; van der Vaart and Wellner, 2010),
 - ▶ *Global maximal inequality*

$$\left(\mathbb{E} \left[\sup_{f \in \mathcal{F}} |\mathbb{G}_n(f)|^q \right] \right)^{1/q} \lesssim \|F\|_{P,q \vee 2} \sqrt{v \log(A \vee n)}, \quad q \in [1, \infty)$$

- ▶ *Local maximal inequality*

$$\mathbb{E} \left[\sup_{f \in \mathcal{F}} |\mathbb{G}_n(f)| \right] \lesssim \sigma \sqrt{v \log(A \vee n)} + \frac{\|\max_{i=1,\dots,n} F(W_i)\|_{P,2}}{\sqrt{n}} v \log(A \vee n).$$

where $\sigma^2 = \sup_{f \in \mathcal{F}} \mathbb{E} f^2$, $\|f\|_{P,q} = (\mathbb{E}|f|^q)^{1/q}$.

- ▶ Both are absent under multiway-cluster dependence.

Representations for Multiway Dependence

- ▶ Suppose the data $\{W_{\mathbf{i}} : \mathbf{i} = (i_1, \dots, i_K) \in \mathbb{N}^K\}$ exhibit K -way dependence
- ▶ Assume that the data r.v.'s are separately exchangeable and dissociated, or equivalently, they admit the *Aldous–Hoover–Kallenberg representation*:

$$W_{\mathbf{i}} \stackrel{d}{=} \tau\left(\{U_{\mathbf{i} \odot \mathbf{e}}\}_{\mathbf{e} \in \{0,1\}^K \setminus \{0\}}\right), \quad (\mathbf{AHK})$$

where $\{U_{\mathbf{i} \odot \mathbf{e}}\}_{\mathbf{e} \in \{0,1\}^K \setminus \{0\}, \mathbf{i} \in [N]}$ are i.i.d. r.v.'s, τ is Borel-measurable, and $\odot$ is the pointwise product⁴.

⁴For $\mathbf{i} = (i_1, \dots, i_K)$, $\mathbf{j} = (j_1, \dots, j_K)$, $\mathbf{i} \odot \mathbf{j} = (i_1 \cdot j_1, \dots, i_K \cdot j_K)$

Example: $K = 2$

- ▶ Suppose $K = 2$, i_1 a market, i_2 a product
- ▶ The AHK representation implies existence of the representation

$$W_{i_1 i_2} \stackrel{d}{=} \tau(U_{i_1 0}, U_{0 i_2}, U_{i_1 i_2}),$$

where

- ▶ $U_{i_1 0}$: market specific shock
- ▶ $U_{0 i_2}$: a product specific shock
- ▶ $U_{i_1 i_2}$: interactive shock

and the shocks are all mutually independent

$\Rightarrow W_{i_1 i_2}$ and $W_{i'_1 i'_2}$ are possibly dependent if either $i_1 = i'_1$ or $i_2 = i'_2$

Empirical Processes with Multiway Dependence

- ▶ Given a data $\{W_{\mathbf{i}} : \mathbf{i} \in I_{\mathbf{N}} := \{1, \dots, N_1\} \times \dots \times \{1, \dots, N_K\}\}$
- ▶ Can accommodate heterogeneous number of observations per $\mathbf{i}$ cell with a little rewriting—theory remains unchanged
- ▶ $n = \min\{N_1, \dots, N_K\}$
- ▶ Denote for f

$$\mathbb{G}_n(f) = \frac{\sqrt{n}}{|I_{\mathbf{N}}|} \sum_{\mathbf{i} \in I_{\mathbf{N}}} (f(W_{\mathbf{i}}) - \mathbb{E}[f(W_{\mathbf{i}})])$$

- ▶ Bounding the whole empirical processes is difficult due to dependence
 $\Rightarrow$ Use orthogonal projections or *Hoeffding-type decomposition* (Hoeffding 1948; Chiang, Kato, Sasaki, 2023)

Multiway Interactions

- ▶ For each $k \in \{1, \dots, K\}$, define

$$\mathcal{E}_k = \left\{ \mathbf{e} = (e_1, \dots, e_K) \in \{0, 1\}^K : \sum_{j=1}^K e_j = k \right\}$$

Interpretation: $\mathcal{E}_k$ consists of all the possible k -way interactions and $\mathbf{e} \in \mathcal{E}_k$ indicates which k -way interactions are turned on

- ▶ $\{0, 1\}^K = \bigcup_{k=0}^K \mathcal{E}_k$
- ▶ Let $\mathbf{e}_k \in \mathcal{E}_1$ be a vector with all zero elements except for the k -th entry.

Example: $K = 2$, i_1 a market, i_2 a product

- ▶ $\mathcal{E}_1 = \{(1, 0), (0, 1)\} = \{e_1, e_2\}$ (market or product-specific effects)
- ▶ $\mathcal{E}_2 = \{(1, 1)\}$ (interactive effect)

Hoeffding-Type Decomposition

- ▶ For $\mathbf{a} = (a_1, \dots, a_K)$, $\mathbf{b} = (b_1, \dots, b_K)$, write $\mathbf{a} \leq \mathbf{b}$ if $a_j \leq b_j$ for all $1 \leq j \leq K$
- ▶ WLOG $\mathbb{E}[f(W_i)] = 0$ and define for each $\mathbf{e} \in \{0, 1\}^K$

$$(P_{\mathbf{e}}f)\left(\{U_{i \odot \mathbf{e}'}\}_{\mathbf{e}' \leq \mathbf{e}}\right) = \mathbb{E}\left[f(W_i) \mid \{U_{i \odot \mathbf{e}'}\}_{\mathbf{e}' \leq \mathbf{e}}\right]$$

- ▶ Hoeffding-type projections: for each unit vector $\mathbf{e} \in \mathcal{E}_1$ that

$$(\pi_{\mathbf{e}}f)(U_{i \odot \mathbf{e}}) = (P_{\mathbf{e}}f)(U_{i \odot \mathbf{e}}),$$

and for each $\mathbf{e} \in \bigcup_{k=2}^K \mathcal{E}_k$ define recursively

$$(\pi_{\mathbf{e}}f)\left(\{U_{i \odot \mathbf{e}'}\}_{\mathbf{e}' \leq \mathbf{e}}\right) = (P_{\mathbf{e}}f)\left(\{U_{i \odot \mathbf{e}'}\}_{\mathbf{e}' \leq \mathbf{e}}\right) - \sum_{\substack{\mathbf{e}' \leq \mathbf{e} \\ \mathbf{e}' \neq \mathbf{e}}} (\pi_{\mathbf{e}'}f)\left(\{U_{i \odot \mathbf{e}''}\}_{\mathbf{e}'' \leq \mathbf{e}'}\right)$$

Example

- ▶ $K = 2$, i_1 a market, i_2 a product
- ▶ Hoeffding-type projections:

$$(\pi_{(1,0)}f)(U_{i_1 0}) = \mathbb{E}[f(W_{i_1 i_2}) \mid U_{i_1 0}] \quad (\text{market-specific effect})$$

$$(\pi_{(0,1)}f)(U_{0 i_2}) = \mathbb{E}[f(W_{i_1 i_2}) \mid U_{0 i_2}] \quad (\text{product-specific effect})$$

$$(\pi_{(1,1)}f)(U_{i_1 0}, U_{0 i_2}, U_{i_1 i_2}) = f(W_{i_1 i_2}) - \mathbb{E}[f(W_{i_1 i_2}) \mid U_{i_1 0}] - \mathbb{E}[f(W_{i_1 i_2}) \mid U_{0 i_2}]$$

(interactive effect)

- ▶ Hence by telescoping

$$\begin{aligned} f(W_{i_1 i_2}) &= (\pi_{(1,0)}f)(U_{i_1 0}) + (\pi_{(0,1)}f)(U_{0 i_2}) + (\pi_{(1,1)}f)(U_{i_1 0}, U_{0 i_2}, U_{i_1 i_2}) \\ &= \sum_{k=1}^2 \sum_{e \in \mathcal{E}_k} (\pi_e f)(\{U_{i \odot e'}\}_{e' \leq e}) \end{aligned}$$

- This yields the Hoeffding-type decomposition

$$\mathbb{G}_n(f) = \sqrt{n} \sum_{k=1}^K \sum_{\mathbf{e} \in \mathcal{E}_k} H_N^{\mathbf{e}}(f),$$

where for each $\mathbf{e} \in \bigcup_{k=1}^K \mathcal{E}_k$, the corresponding *Hoeffding-type projection*

$$H_N^{\mathbf{e}}(f) = \frac{1}{|I_{N,\mathbf{e}}|} \sum_{\mathbf{i} \in I_{N,\mathbf{e}}} (\pi_{\mathbf{e}} f)(\{U_{\mathbf{i} \odot \mathbf{e}'}\}_{\mathbf{e}' \leq \mathbf{e}}),$$

$$I_{N,\mathbf{e}} = \{\mathbf{i} \odot \mathbf{e} : \mathbf{i} \in I_N\}$$

$\Rightarrow$ To control $\mathbb{G}_n(f)$, it suffices to control $H_N^{\mathbf{e}}(f)$ for each $\mathbf{e} \in \{0, 1\}^K$

Local Maximal Inequality

Theorem For each $\mathbf{e} \in \mathcal{E}_k$, let $\sigma_{\mathbf{e}}$ be a constant such that

$$\sup_{f \in \mathcal{F}} \|P_{\mathbf{e}} f\|_{P,2} \leq \sigma_{\mathbf{e}} \leq \|P_{\mathbf{e}} F\|_{P,2}, \quad M_{\mathbf{e}} = \max_{t=1,\dots,n} P_{\mathbf{e}} F(\{U_{(t,\dots,t)} \odot \mathbf{e}'\} | \mathbf{e}' \leq \mathbf{e}).$$

If $\mathcal{F}$ is VC-type,

$$\begin{aligned} & |I_{\mathbf{N},\mathbf{e}}|^{1/2} \mathbb{E} \left[\sup_{f \in \mathcal{F}} |H_{\mathbf{N}}^{\mathbf{e}}(f)| \right] \\ & \lesssim \sigma_{\mathbf{e}} \{v \log(A \|P_{\mathbf{e}} F\|_{P,2} / \sigma_{\mathbf{e}})\}^{k/2} + \frac{\|M_{\mathbf{e}}\|_{P,2}}{\sqrt{n}} \{v \log(A \|P_{\mathbf{e}} F\|_{P,2} / \sigma_{\mathbf{e}})\}^k \end{aligned}$$

- ▶ The first term: $\sigma_{\mathbf{e}}$ ("Size") typically decreasing as the $v \log(\dots)$ ("Complexity") increases
- ▶ The second term: often negligible due to the extra $1/\sqrt{n}$
- ▶ Can be used to show $\sqrt{n}|III| = o_p(1)$ and thus DML without cross-fitting

Learning Nuisance Parameter under Multiway Dependence

- ▶ Our maximal inequalities are also useful for establishing convergence rate $\|\hat{\eta} - \eta_0\|_{P,2}$ under multiway dependence
- ▶ e.g. for LASSO with a p -dimensional predictor, we need a bound in probability for

$$\max_{1 \leq j \leq p} \left| \sum_{\mathbf{i} \in I_{\mathbf{N}}} X_{\mathbf{i},j} \varepsilon_{\mathbf{i}} \right|$$

- ▶ Under multiway dependence, this can again be controlled by using maximal inequality + Hoeffding-type decomposition

Remarks on the Proofs

- ▶ The proofs of both maximal inequalities rely on symmetrisation (Chiang, Kato, Sasaki, 2023), the maximal inequality for Rademacher chaos (de la Peña and Giné, 1999), and, in the case of local maximal inequality, Hoffmann–Jørgensen inequality
- ▶ The related maximal inequality for U -processes is obtained using the classical Hoeffding-averaging technique to allow for an application of Hoffmann–Jørgensen m
 - ⇒ Hoeffding-averaging is not applicable for multiway means
- ▶ To overcome this difficulty, we show that each $\{\mathbf{i} \odot \mathbf{e} : \mathbf{i} \in [\mathbf{N}]\}$ can be partitioned into groups such that the sum within each group consists of n i.i.d. random variables, allowing the application of Hoffmann–Jørgensen

Summary

- ▶ Existing DML methods rely on **cross-fitting**, but under multiway dependence it is **complicated** and **causes extra issues**.
- ▶ To study DML methods without cross-fitting, we first develop **maximal inequalities under multiway-cluster dependence**.
 - ▶ Also of independent interest.
- ▶ Applying our results, we characterize the complexity conditions of the score class indexed by the nuisance parameter, under which **DML without cross-fitting is valid**.
 - ▶ We verify that common machine learner satisfy such complexity conditions.
- ▶ In simulation, we illustrate the **issue with cross-fitted DML** as well as the **substantial computational advantage** with DML without cross-fitting.

Thank you!

Questions and comments welcome.